Measure Theoretic Probability
Lesson 19
Combining Random Variables and Almost Sure Convergence

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Outline

1. Recap: Measurable Functions

2. New Random Variables from Old

3. Almost Sure Convergence

4. Summary and Exercises

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Measurable Functions and Random Variables

Measurable function

Let $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ be measurable spaces. A function $f : \Omega_1 \to \Omega_2$ is **measurable** if $f^{-1}(B) \in \mathcal{F}_1$ for every $B \in \mathcal{F}_2$. When $\Omega_2 = \mathbb{R}$ and no σ-field is specified, $\mathcal{F}_2 = \mathcal{B}(\mathbb{R})$.

Random variable

A random variable is a measurable function $X : \Omega \to \mathbb{R}$. No measure is involved in this condition, so everything today holds for **any** real-valued measurable functions, probability or not.

Why this lesson

Before treating almost sure convergence of a sequence X_n, we must know that sums, scalar multiples and absolute values of random variables are again random variables, so that sets such as $\{|X_n - X| < \varepsilon\}$ are events.

Rays Criterion, with Other Generating Classes

Theorem

$X : \Omega \to \mathbb{R}$ is a random variable $\iff$ $\{X \le x\} = X^{-1}\big((-\infty, x]\big) \in \mathcal{F}$ for every $x \in \mathbb{R}$.

Why other sets work too

($\Rightarrow$) is just measurability, since the set is Borel. ($\Leftarrow$) uses only that these sets **generate** $\mathcal{B}(\mathbb{R})$. Hence we may replace $(-\infty, x]$ by any generating class, for example

$$\{X \ge x\}, \quad \{X > x\}, \quad \{X < x\}, \quad \{x < X < y\} \quad (x < y).$$

Practical consequence

To show X is a random variable, check any one of these families lies in $\mathcal{F}$, instead of checking every Borel set.

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Lemma 1: $\{X > Y\} \in \mathcal{F}$

Claim

Let X, Y be random variables on $(\Omega, \mathcal{F})$. Then $\{\omega : \ X(\omega) > Y(\omega)\} \in \mathcal{F}$.

Proof

Separate X and Y by a rational number:

$$\{X > Y\} = \bigcup_{r \in \mathbb{Q}} \big(\{X > r\} \cap \{r > Y\}\big).$$

$(\supseteq)$ is clear. $(\subseteq)$ holds because the rationals are **dense**: if $X(\omega) > Y(\omega)$, some rational r lies strictly between them.

$\{X > r\} = X^{-1}\big((r, \infty)\big) \in \mathcal{F}$ and $\{Y < r\} = Y^{-1}\big((-\infty, r)\big) \in \mathcal{F}$. A σ-field is closed under finite intersections, and $\mathbb{Q}$ is **countable**, so the countable union lies in $\mathcal{F}$. □

Remarks

A union over all of $\mathbb{R}$ would not be allowed, since $\mathbb{R}$ is uncountable. The same proof works for $<$, $\geq$, $\leq$, and (with a little more care) for $=$.

Lemma 2: $X + a$ and aX

$X + a$ is a random variable ($a \in \mathbb{R}$)

For $x \in \mathbb{R}$,

$$\{X + a \leq x\} = \{X \leq x - a\} = X^{-1}\big((-\infty, x - a]\big) \in \mathcal{F}.$$

aX is a random variable ($a \in \mathbb{R}$)

Fix $x \in \mathbb{R}$ and consider $\{aX \leq x\}$.

- $a > 0$: $\{aX \leq x\} = \{X \leq x/a\} \in \mathcal{F}$.
- $a = 0$: $\{0 \leq x\}$ is Ω if $x \geq 0$ and $\varnothing$ if $x < 0$; both lie in $\mathcal{F}$.
- $a < 0$: dividing by a flips the inequality, $\{aX \leq x\} = \{X \geq x/a\} = X^{-1}\big([x/a, \infty)\big) \in \mathcal{F}$.

Remark

The case $a = -1$ gives: if Y is a random variable, so is $-Y$.

Theorem: $X + Y$ Is a Random Variable

Theorem

If X, Y are random variables on $(\Omega, \mathcal{F})$, then $X + Y$ is a random variable. Equivalently, the sum of measurable functions is measurable.

Proof

Fix $x \in \mathbb{R}$. Move Y to the right-hand side:

$$\{X + Y \leq x\} = \{X \leq x - Y\}.$$

By Lemma 2, $Z := x - Y = (-1)Y + x$ is a random variable. The complement of this set is $\{X > Z\}$, which lies in $\mathcal{F}$ by Lemma 1. Since $\mathcal{F}$ is closed under complements, $\{X + Y \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$. By the rays criterion, $X + Y$ is a random variable. □

Consequence

$X - Y = X + (-1)Y$ is also a random variable. This is exactly what we need for $X_n - X$.

$|X|$ Is a Random Variable

Claim

If X is a random variable, so is $|X|$.

Proof

Fix $x \in \mathbb{R}$ and look at $\{|X| \leq x\}$.

- If $x < 0$: no ω has $|X(\omega)| \leq x$, so the set is $\varnothing \in \mathcal{F}$.
- If $x \geq 0$: $\{|X| \leq x\} = \{-x \leq X \leq x\} = X^{-1}\big([-x, x]\big) \in \mathcal{F}$, since $[-x, x]$ is a Borel set.

In both cases the set lies in $\mathcal{F}$, so $|X|$ is a random variable. □

Remarks

Closed intervals $[-x, x]$ alone do not generate $\mathcal{B}(\mathbb{R})$, but that is irrelevant here: we only need the set to be Borel and X to be measurable. This is a special case of the general fact that a composition of measurable functions is measurable.

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Definition

Setting

$X_1, X_2, \ldots$ and X are random variables on a **common** probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition

X_n **converges almost surely** to X if

$$\mathbb{P}\Big(\big\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\big\}\Big) = 1,$$

written $X_n \xrightarrow{\text{a.s.}} X$. Equivalent names: **convergence with probability one**, and **almost everywhere** convergence in general measure theory.

A point to settle

For $\mathbb{P}$ of this set to make sense, the set must belong to $\mathcal{F}$. We prove this at the end of the lesson.

Example: Setup

Probability space

$\Omega = [0,1]$, $\mathcal{F} = \mathcal{B}([0,1])$, $\mathbb{P}$ = Lebesgue measure (length).

The sequence and the candidate limit

$$X_n(\omega) = \begin{cases} 1, & 0 \le \omega < \dfrac{2n+1}{3n}, \\ 0, & \text{otherwise}, \end{cases} \qquad X(\omega) = \begin{cases} 1, & 0 \le \omega < \dfrac{2}{3}, \\ 0, & \text{otherwise}. \end{cases}$$

Observation

$\dfrac{2n+1}{3n} = \dfrac{2}{3} + \dfrac{1}{3n}$ decreases to $\dfrac{2}{3}$ **from above**. So some $\omega > \frac{2}{3}$ have $X_n(\omega) = 1$ for small n while $X(\omega) = 0$. We check whether this is a problem on a set of probability zero.

Example: Checking Convergence Pointwise

Let $A = \{\omega : \lim_n X_n(\omega) = X(\omega)\}$. Three cases.

- $0 \le \omega < \frac{2}{3}$: $X_n(\omega) = 1$ for all n and $X(\omega) = 1$. Constant sequence, so $X_n(\omega) \to X(\omega)$.
- $\frac{2}{3} < \omega \le 1$: $X(\omega) = 0$. Now $X_n(\omega) = 1 \iff \omega < \frac{2n+1}{3n} \iff n(3\omega - 2) < 1 \iff n < \dfrac{1}{3\omega - 2}$.
  So $X_n(\omega) = 0$ for all $n \ge \dfrac{1}{3\omega - 2}$, and $X_n(\omega) \to 0 = X(\omega)$.
- $\omega = \frac{2}{3}$: $X_n(\frac{2}{3}) = 1$ for every n (since $\frac{2}{3} < \frac{2}{3} + \frac{1}{3n}$), but $X(\frac{2}{3}) = 0$. No convergence.

Conclusion

$A = \left[0, \frac{2}{3}\right) \cup \left(\frac{2}{3}, 1\right]$, which is $[0, 1]$ with one point removed. Hence $\mathbb{P}(A) = 1 - 0 = 1$, so

$$X_n \xrightarrow{\text{a.s.}} X.$$

Why A Is an Event: Rewriting A with Quantifiers

Claim

For random variables X_n, X, the set $A = \{\omega : \ X_n(\omega) \to X(\omega)\}$ lies in $\mathcal{F}$.

Step 1: unwind the definition

For fixed ω, $X_n(\omega) \to X(\omega)$ is a statement about real numbers:

$$\forall\, \varepsilon > 0 \ \ \exists\, N \ \ \forall\, n \geq N : \ \ |X_n(\omega) - X(\omega)| < \varepsilon.$$

In set language, "for all" becomes $\bigcap$ and "there exists" becomes $\bigcup$:

$$A = \bigcap_{\varepsilon > 0} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \{|X_n - X| < \varepsilon\}.$$

Problem

The outer intersection is over **uncountably many** ε, but a σ-field is closed only under **countable** intersections.

Why A Is an Event: The Countable Trick

Step 2: replace ε by $1/k$

For every $\varepsilon > 0$ there is a positive integer k with $1/k \le \varepsilon$. So the condition for all ε is equivalent to the condition for all $\varepsilon = 1/k$:

$$\omega \in A \iff \forall k \in \mathbb{N}\ \exists N\ \forall n \ge N :\ |X_n(\omega) - X(\omega)| < \tfrac{1}{k}.$$

Therefore

$$A = \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left\{ |X_n - X| < \tfrac{1}{k} \right\}.$$

All three operations are now **countable**.

Step 3: the innermost sets are events

$X_n - X$ is a random variable (sum theorem with $a = -1$), $|X_n - X|$ is a random variable ($|\cdot|$ result), and

$$\left\{ |X_n - X| < \tfrac{1}{k} \right\} = |X_n - X|^{-1}\big((-\infty, \tfrac{1}{k})\big) \in \mathcal{F}.$$

$\mathcal{F}$ is closed under countable intersections and unions, so $A \in \mathcal{F}$. □

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Summary

Key results

Concept / Result	Content
Rays criterion	any generating class of sets may be pulled back, e.g. $\{X < x\}$, $\{X > x\}$, $\{x < X < y\}$
Lemma 1	$\{X > Y\} = \bigcup_{r \in \mathbb{Q}} \{X > r\} \cap \{Y < r\} \in \mathcal{F}$
Lemma 2	$X + a$ and aX are random variables
Sum	$X + Y$ is a random variable (via $\{X \leq x - Y\}$)
Absolute value	$\lvert X \rvert$ is a random variable
Almost sure convergence	$\mathbb{P}\big(\lim_n X_n = X\big) = 1$
Measurability of the limit set	$A = \bigcap_k \bigcup_N \bigcap_{n \geq N} \{\lvert X_n - X \rvert < 1/k\} \in \mathcal{F}$

Exercises

1. Prove that $\{X \geq Y\} \in \mathcal{F}$ and $\{X = Y\} \in \mathcal{F}$ for random variables X, Y.
2. Show that X is a random variable iff $\{X \geq x\} \in \mathcal{F}$ for all $x \in \mathbb{R}$.
3. Show that if X, Y are random variables then so is $X - Y$.
4. Show that $\max(X, Y)$ is a random variable. (Hint: $\{\max(X, Y) \leq x\} = \{X \leq x\} \cap \{Y \leq x\}$.)
5. In the example, verify that $X_n(\frac{2}{3}) = 1$ for all n, and compute $\mathbb{P}(A)$.
6. Redo the example with $X_n = I_{[0,\, 2/3 - 1/(3n))}$. Does $X_n \xrightarrow{\text{a.s.}} X$? Describe A.
7. Show that $A = \bigcap_{\varepsilon > 0} \bigcup_N \bigcap_{n \geq N} \{|X_n - X| < \varepsilon\}$ is the same as the version with $\varepsilon = 1/k$.

Outlook

Next lesson

More concrete examples of almost sure convergence, and an **equivalent criterion** for almost sure convergence that is often easier to apply than the definition.

Looking ahead

Open question: if $X_n \to X$ pointwise (or almost surely) and the X_n are random variables, is the limit X itself a random variable? We also define convergence in probability and show that almost sure convergence implies it, but not conversely.